



THE KING'S SCHOOL

2003 Higher School Certificate Trial Examination

Mathematics

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question

Total marks – 120

- Attempt Questions 1-10
- All questions are of equal value

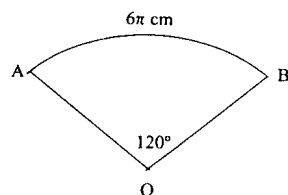
Total marks – 120
Attempt Questions 1-10
All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Marks

Question 1 (12 marks) Use a SEPARATE writing booklet.

- (a) Find, correct to two decimal places, $\log_{12} 2003$ 2
- (b) Find the derivative of $12 - \cos 12x$ 2
- (c) In sector OAB , $\angle AOB = 120^\circ$ and arc $AB = 6\pi$ cm. Find the radius of the sector. 2

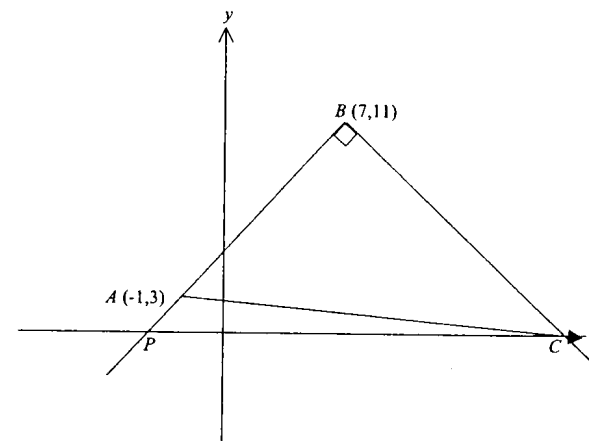


- (d) Solve $\sqrt{2}x = 2\sqrt{3}$, expressing your solution in simplest form. 2
- (e) Sketch the region in the number plane where $0 \leq y \leq e^{-x}$ 2
- (f) Simplify $\frac{x^2}{x + \frac{x}{x-1}}$ 2

Marks

Question 2 (12 marks) Use a SEPARATE writing booklet.

(a)



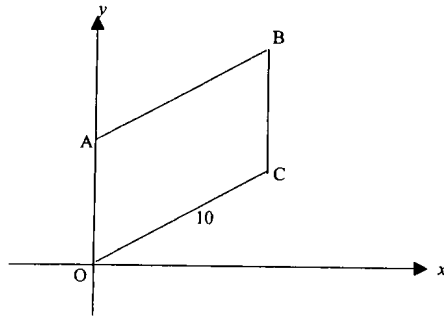
In the diagram, PAB is a straight line where P is on the x axis.

$\triangle ABC$ has vertices $A(-1, 3)$, $B(7, 11)$ and C , which is on the x axis. $\angle ABC = 90^\circ$.

- (i) Find the size of $\angle BPC$ 2
- (ii) Find the equation of BC 2
- (iii) State the coordinates of point C 1
- (iv) Find the area of $\triangle ABC$ 2
- (v) Find the size of $\angle BAC$, nearest degree 2

Question 2 (continued)

(b)



OABC is a parallelogram, O is the origin and A is on the y axis.
The equations of OC and AB are $y = 2x$ and $y = 2x + 5$, respectively.
The length of OC is 10 units.

Find the area of the parallelogram.

3

Question 3 (12 marks) Use a SEPARATE writing booklet.

(a) Find, correct to 1 decimal place,

$$\int_0^{0.1} \sec^2(x+1) dx$$

2

(b) The probability that Max can correctly integrate a function is 0.7. Max is given 7 functions to integrate.

Find the probability that Max gets at least one integration wrong. Give your answer correct to 1 decimal place.

2

(c) Find the sum of the arithmetic series

$$-7 + (-2) + 3 + \dots + 2003$$

3

(d) (i) Sketch on the same diagram

$$y = |x - 2| \text{ and } y = 2x,$$

showing the x and y intercepts.

2

(ii) Hence, or otherwise,

$$\text{solve } |x - 2| = 2x$$

2

(iii) Using (i), or otherwise,

$$\text{find } \int_0^4 |x - 2| dx$$

1

Marks

Question 4 (12 marks) Use a SEPARATE writing booklet.

- (a) Find the equation of the tangent to the curve $y = (x+1)e^x$ at the point where $x = 0$

3

- (b) Prove that $\sec^2 A + \operatorname{cosec}^2 A \equiv \sec^2 A \operatorname{cosec}^2 A$

3

- (c) Find the centre and radius of the circle $x^2 + y^2 = 4y$

3

- (d) The line $y = 2x + c$ is a tangent to the parabola $y = x^2 + x$.

Find the value of c .

3

Marks

Question 5 (12 marks) Use a SEPARATE writing booklet.

- (a) Consider the curve $y = x^3(x-4)$

- (i) State the x intercepts.

1

- (ii) Show that there are stationary points at $x = 0$ and $x = 3$

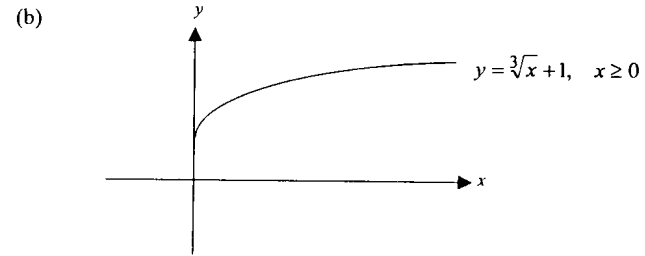
2

- (iii) Show that the stationary point at $x = 0$ is a point of inflection.

2

- (iv) Sketch the curve.

2



The diagram shows the sketch of $y = \sqrt[3]{x} + 1$ for $x \geq 0$

- (i) Show that the line $y = \frac{1}{4}x + 1$ meets the curve $y = \sqrt[3]{x} + 1, \quad x \geq 0$, at $x = 0, \quad 8$

1

- (ii) Copy the diagram into your booklet and include on it the line $y = \frac{1}{4}x + 1$

1

- (iii) Find the area enclosed between the line and the curve on your diagram.

3

Question 6 (12 marks) Use a SEPARATE writing booklet.

- (a) A quantity Q is decreasing at the rate $\frac{dQ}{dt} = kQ$, k a constant.

Q is in grams and t is time measured in hours.

Initially, $Q = 30$ and 3 hours later, $Q = 9$

- (i) Show that $Q = 30e^{kt}$ satisfies both the initial condition and the equation $\frac{dQ}{dt} = kQ$

2

- (ii) Find the one significant figure value for k .

2

- (iii) How much of the quantity, correct to one significant figure, will be left after a further one hour has elapsed?

2

- (b) From P , a ship sails on a bearing of 070° to A , a distance of 150 km. Also, from P , another ship sails on a bearing of 330° to B , a distance of 300 km.

- (i) Draw a diagram to show the above information.

1

- (ii) Find the distance from A to B , correct to the nearest kilometre.

2

- (iii) Find the bearing of B from A , correct to the nearest degree.

3

Question 7 (12 marks) Use a SEPARATE writing booklet.

- (a) Solve the equation $(3x-1)^4 - 2(3x-1)^2 - 8 = 0$

4

- (b) A bag contains six discs. Two of the discs have the number 0 on them and the other four discs have the number 1 on them.

Three discs are withdrawn at random.

- (i) Find the probability that all of the three discs drawn have the number 1 on them.

2

- (ii) Find the probability that the product of the numbers on the three discs drawn is 0.

1

- (c) Maggie borrows \$10 000 from a bank. This loan plus interest and charges are to be repaid at the end of each month in equal monthly instalments, \$ M , over five years. Interest of 12% p.a. on the balance owing at the start of each month is added to the account at the end of each month. Additionally, at the end of each month a management charge of \$10 is added to the account.

Let \$ A_n be the amount owing after n months.

- (i) Show that $A_1 = 10\,000 \times 1.01 - (M - 10)$

1

- (ii) Show that $A_2 = 10\,000 \times 1.01^2 - (M - 10)(1 + 1.01)$

1

- (iii) Find \$ M , correct to the nearest cent.

3

Question 8 (12 marks) Use a SEPARATE writing booklet.

(a) Find

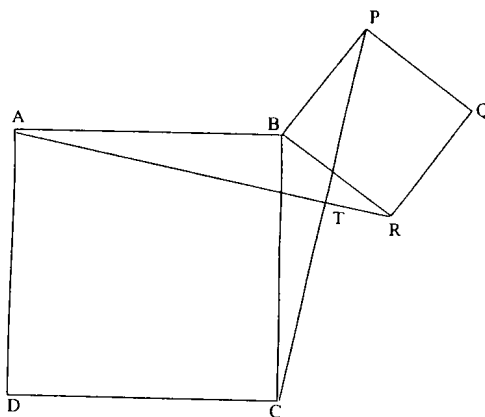
(i) $\int \frac{4x^4}{4x^5 + 1} dx$

2

(ii) $\int \frac{4x^5 + 1}{4x^4} dx$

3

(b) In the diagram, ABCD and BPQR are squares. AR intersects PC at T.



(i) Copy the diagram into your booklet.

(ii) Prove $\triangle ABR \cong \triangle CBP$

3

(iii) Why does $\angle BAR = \angle PCB$?

1

(iv) Prove that $AR \perp PC$

3

Question 9 (12 marks) Use a SEPARATE writing booklet.

(a) (i) Sketch the curve $y = 2\sin\left(\frac{x}{2}\right) + 1$, $0 \leq x \leq 3\pi$

2

(ii) The region bounded by the curve in (i) and the x axis from $x = 0$ to $x = \pi$ is revolved about the x axis.

Write down a definite integral which would give the volume of the solid of revolution.

1

(iii) Use Simpson's Rule with 3 function values to give a one decimal place approximation to the volume in (ii).

3

(b) A particle moves on the x axis with its velocity, v m/s, given at any time, t seconds, $t \geq 0$, by $v = \frac{1}{\sqrt{2t+1}}$

Initially the particle is at the origin.

(i) Find the initial velocity and the velocity after 12 seconds.

1

(ii) Sketch the velocity-time graph.

1

(iii) Find the acceleration of the particle after 12 seconds.

2

(iv) Find the displacement of the particle as a function of time.

2

Marks

Question 10 (12 marks) Use a SEPARATE writing booklet.

- (a) Find the equation of the directrix of the parabola $(x+1)^2 = 4y+2$ 2
- (b) A circle and two equal squares are to have a total perimeter of 200 cm.
Let the radius of the circle be $4x$ cm.
- (i) Show that each side of the squares is $25 - \pi x$ cm. 2
- (ii) Deduce that $0 \leq x \leq \frac{25}{\pi}$ 1
- (iii) Show that the total area, $A \text{ cm}^2$, of the circle and the two squares is given by
- $$A = 2\pi(8 + \pi)x^2 - 100\pi x + 1250$$
- 2
- (iv) Find the exact value for x for which A is a minimum. 3
- (v) Find the exact value of the minimum area in simplest form. 1
- (vi) Find the exact value for x for which A is a maximum. Give reasons. 1

End of Paper

Mathematics

Qn 1

$$(a) \log_{12} 2003 = \frac{\log_{10} 2003}{\log_{10} 12} \text{ or } \frac{\ln 2003}{\ln 12} = \boxed{3.06}$$

$$(b) \frac{d}{dx}(12 - \cos 12x) = -(-\sin 12x) \times 12 = \boxed{12 \sin 12x}$$

$$(c) 120^\circ = \frac{2\pi}{3} \text{ radians, } l = r\theta$$

$$\therefore \frac{2\pi}{3} \times r = 6\pi$$

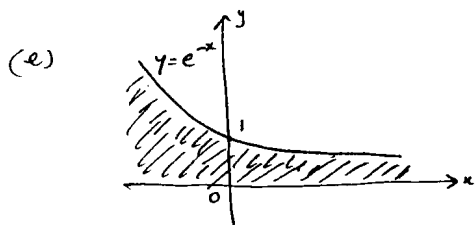
$$\Rightarrow r = 6\pi \times \frac{3}{2\pi} = 9$$

i.e. radius is $\boxed{9 \text{ cm}}$

OR, alternatively, $120^\circ = \frac{1}{3} \times 360^\circ$

$$\Rightarrow \frac{1}{3} \times 2\pi r = 6\pi \quad \therefore r = 9$$

$$(d) x = \frac{2\sqrt{3}}{\sqrt{2}} = \sqrt{2}\sqrt{3} = \boxed{\sqrt{6}}$$



$$(f) \frac{x^2}{x + \frac{x}{x-1}} = \frac{x^2(x-1)}{x(x-1) + x}$$

$$= \frac{x^2(x-1)}{x^2 - x + x}$$

$$= \frac{x^2(x-1)}{x^2} = \boxed{x-1}$$

Qn 2

$$(a) (i) \text{ gradient of } AB = \frac{11-3}{7-1} = \frac{8}{6} = 1$$

$$\Rightarrow \tan \angle BAC = 1 \quad \therefore \boxed{\angle BAC = 45^\circ}$$

$$(ii) \text{ gradient of } BC = -1 \quad [BC \perp AB]$$

$$\therefore BC \text{ is } y-11 = -(x-7) = -x+7$$

$$\text{i.e. } \boxed{y = -x + 18}$$

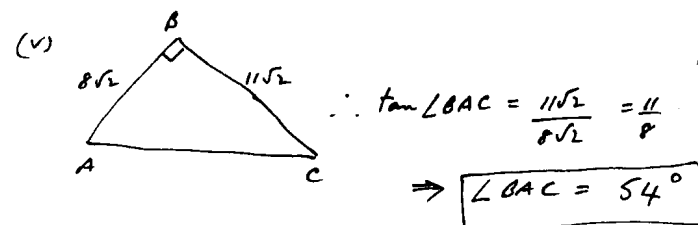
$$(iii) \text{ at } C, y=0 \Rightarrow -x+18=0 \quad \therefore \boxed{C=(18, 0)}$$

$$(iv) AB = \sqrt{(7-1)^2 + (11-3)^2} = \sqrt{8^2 + 8^2} = 8\sqrt{2}$$

$$BC = \sqrt{(18-7)^2 + (0-11)^2} = \sqrt{11^2 + 11^2} = 11\sqrt{2}$$

$$\therefore \text{Area} = \frac{1}{2} \cdot 8\sqrt{2} \cdot 11\sqrt{2} \text{ u}^2 = \boxed{88 \text{ u}^2}$$

(There are alternatives, of course)



$$(b) AB \text{ is } 2x - y + 5 = 0$$

$\therefore$ perpendicular distance from $(0, 0)$ to AB

$$= \frac{0-0+5}{\sqrt{2^2+1^2}} = \frac{5}{\sqrt{5}} = \sqrt{5}$$

$$\therefore \text{Area} = 10 \times \sqrt{5} \text{ u}^2 = \boxed{10\sqrt{5} \text{ units}^2}$$

[There are many alternatives]

Qn 3

$$(a) \int_0^{0.1} \sec^2(x+1) dx = [\tan(x+1)]_0^{0.1} = \tan 1.1 - \tan 1 = \boxed{0.4}$$

$$(b) P(\text{at least 1 wrong}) = 1 - P(\text{none wrong}) = 1 - (0.7)^7 = \boxed{0.9}$$

$$(c) \text{ In usual notations, } a = -7, d = -2 - (-7) = 5$$

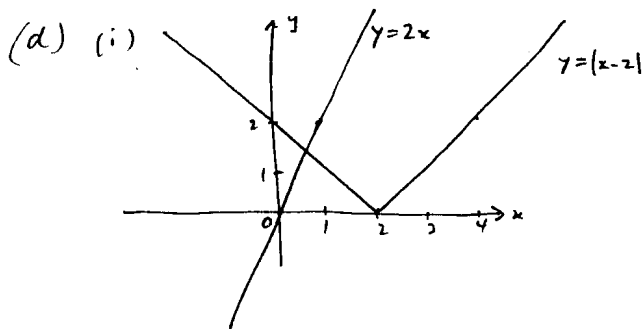
$$\Rightarrow -7 + (n-1)5 = 2003$$

$$5(n-1) = 2010$$

$$n-1 = 402$$

$$\therefore n = 403, \text{ the number of terms}$$

$$\therefore -7 + \dots + 2003 = \frac{403}{2} (-7 + 2003) = \boxed{402194}$$



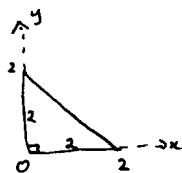
$$(ii) \text{ The diagram indicates } 2x = -(x-2)$$

$$\text{ie } 2x = -x + 2$$

$$\therefore 3x = 2$$

$$\text{or, } \boxed{x = \frac{2}{3}}$$

$$(iii) \text{ The diagram indicates } \int_0^4 |x-2| dx \text{ is twice } 2 \times \frac{1}{2} \cdot 2 \cdot 2 = \boxed{4}$$



Qn 4

$$(a) \frac{dy}{dx} = (x+1)e^x + e^x(1) = e^x(x+2)$$

$$\text{at } x=0, y=1, \frac{dy}{dx} = 2$$

$$\therefore \text{ tangent is } y-1 = 2(x-0) \text{ ie } \boxed{y = 2x+1}$$

$$(b) \sec^2 A + \operatorname{cosec}^2 A = \frac{1}{\cos^2 A} + \frac{1}{\sin^2 A} = \frac{\sin^2 A + \cos^2 A}{\cos^2 A \sin^2 A} = \frac{1}{\cos^2 A \sin^2 A} = \sec^2 A \operatorname{cosec}^2 A$$

$$(c) \text{ Rewrite as } x^2 + y^2 - 4y = 0$$

$$\therefore x^2 + (y-2)^2 - 4 = 0$$

$$\text{or } x^2 + (y-2)^2 = 4$$

$$\therefore \boxed{\text{Centre} = (0, 2), \text{ radius is } 2}$$

$$(d) \text{ Solving simultaneously,}$$

$$x^2 + x = 2x + c$$

$$\text{ie } x^2 - x - c = 0$$

$$\text{For the line to be a tangent we need } \Delta = 0$$

$$\Delta = 1 - 4(-c) = 1 + 4c = 0 \text{ if } \boxed{c = -\frac{1}{4}}$$

$$\text{Alternatively, for } y = x^2 + x$$

$$\frac{dy}{dx} = 2x + 1$$

$$\text{For } y = 2x + c, \text{ the gradient is } 2$$

$$\therefore \text{ we need } 2x + 1 = 2$$

$$\Rightarrow x = \frac{1}{2}$$

$$\text{and } y = \left(\frac{1}{2}\right)^2 + \frac{1}{2} = \frac{3}{4}$$

$$\therefore \text{ for line to be a tangent,}$$

$$\frac{3}{4} = 2 \times \frac{1}{2} + c \Rightarrow c = -\frac{1}{4}$$

Ques 5

(a) (i) $x = 0, 4$

(ii) $y = x^4 - 4x^3$

$$\therefore \frac{dy}{dx} = 4x^3 - 12x^2 = 4x^2(x-3)$$

$$= 0 \text{ if } x = 0, 3$$

i.e. there are stationary points at $x = 0, 3$

(iii) $\frac{d^2y}{dx^2} = 12x^2 - 24x = 12x(x-2)$

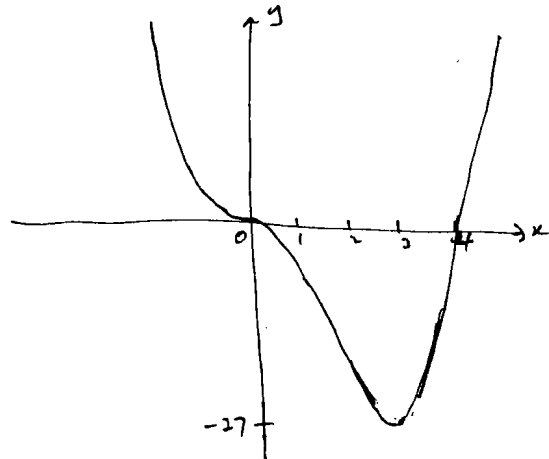
$$\text{If } x = 0, \frac{d^2y}{dx^2} = 0$$

$$\left. \begin{array}{l} \text{and if } x = -1, \frac{d^2y}{dx^2} = -12(-1) > 0 \\ \text{if } x = 1, \frac{d^2y}{dx^2} = 12(-1) < 0 \end{array} \right\} \Rightarrow \text{a change in concavity}$$

$\therefore$ at $x = 0$ there is a (horizontal) point of inflection

(iv) If $x = 3, y = 27(-1) = -27$

$\therefore$ combining (i), (ii), (iii) we must have

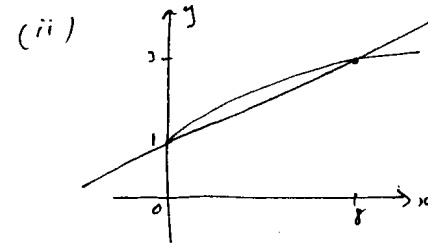


Ques 5

(b) (i) For the line if $x = 0, 8$ then $y = 1, 3$

For the curve if $x = 0, 8$ then $y = 1, \sqrt[3]{8} + 1 = 3$

$\therefore$ result



(iii) $A = \int_0^8 \sqrt[3]{x} + 1 - \left(\frac{1}{4}x + 1\right) dx$

$$= \int_0^8 x^{\frac{1}{3}} - \frac{1}{4}x \, dx$$

$$= \left[\frac{3x^{\frac{4}{3}}}{4} - \frac{1}{4} \cdot \frac{x^2}{2} \right]_0^8$$

$$= \frac{3}{4} \cdot 16 - \frac{1}{8} \cdot 64 - (0)$$

$$= \boxed{4 \text{ u}^2}$$

Q6

(a) (i) if $t=0$, $Q = 30e^0 = 30$ $\therefore$ initial condition satisfied

Next, $\frac{dQ}{dt} = 30k e^{kt} = k(30e^{kt}) = kQ$

$\therefore$ equation $\frac{dQ}{dt} = kQ$ is satisfied

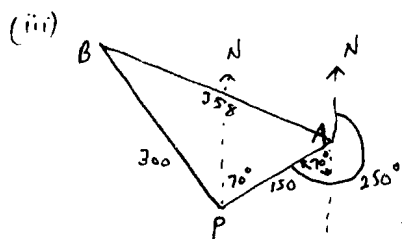
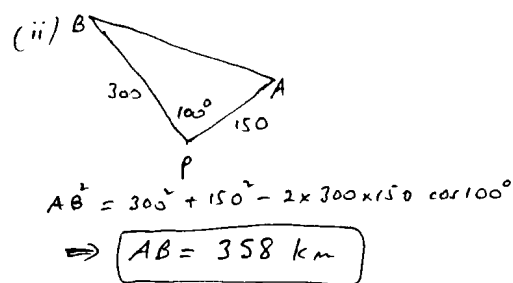
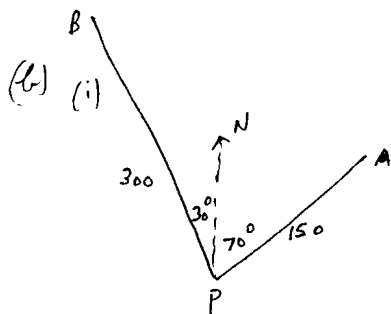
(ii) $t=3$, $Q=9 \Rightarrow 9 = 30e^{3k}$

or $e^{3k} = \frac{9}{30} = 0.3$

$\therefore 3k = \ln 0.3$

or $k = \frac{1}{3} \ln 0.3 = \boxed{-0.4}$

(iii) when $t=4$, $Q = 30e^{-0.4 \times 4} = 30e^{-1.6}$
 $= \boxed{6g}$



In $\triangle PAB$,
 $\cos A = \frac{150^2 + 358^2 - 300^2}{2 \cdot 150 \cdot 358}$
 $\Rightarrow \hat{A} = 56^\circ$, nearest degree

$\therefore$ bearing of B from A is
 $250^\circ + 56^\circ = \boxed{306^\circ}$

Q7

(a) Put $u = (3x-1)^2$

Then, $u^2 - 2u - 8 = 0$

$(u+2)(u-4) = 0$

$\Rightarrow u = -2, 4$

$\therefore (3x-1)^2 = -2$ or 4

* So, $(3x-1)^2 = 4$ only since $(3x-1)^2 \geq 0$

$\therefore 3x-1 = 2$ or -2

Thus, $\boxed{x = 1 \text{ or } -\frac{1}{3}}$

(b) (i) $P(111) = \frac{4}{6} \times \frac{3}{5} \times \frac{2}{4} = \boxed{\frac{1}{5}}$

(ii) The product will be 0 unless all the dice are 1

$\therefore P(\text{zero product}) = 1 - \frac{1}{5} = \boxed{\frac{4}{5}}$

(c) (i) $12\% \text{ p.a.} = 1\% \text{ p.m.} = 0.01$

$\therefore A_1 = 10000 + 0.01 \times 10000 - M + 10$
 $= 10000 \times 1.01 - (M - 10)$

(ii) $A_2 = A_1 \times 1.01 - M + 10$

$= 10000 \times 1.01^2 - (M - 10) \cdot 1.01 - (M - 10)$
 $= 10000 \times 1.01^2 - (M - 10)(1 + 1.01)$

(iii) From (i), (ii),

$A_n = 10000 \times 1.01^n - (M - 10)(1 + 1.01 + \dots + 1.01^{n-1})$

* if $n = 5 \times 12 = 60$, $A_n = 0$

$\therefore (M - 10)(1 + 1.01 + \dots + 1.01^{59}) = 10000 \times 1.01^{60}$

or $(M - 10)(1.01^{60} - 1) = 10000 \times 1.01^{60}$

$\therefore M - 10 = \frac{10000 \times 1.01^{60} \times 0.01}{1.01^{60} - 1} \Rightarrow \boxed{\$M = \$232.44}$

Qn 8

$$(a) (i) \int \frac{4x^4}{4x^5+1} dx = \frac{1}{5} \int \frac{20x^4}{4x^5+1} dx = \boxed{\frac{1}{5} \ln(4x^5+1) + c}$$

$$(ii) \int \frac{4x^5+1}{4x^4} dx = \int \frac{4x^5}{4x^4} + \frac{1}{4x^4} dx$$

$$= \int x + \frac{1}{4} x^{-4} dx$$

$$= \frac{x^2}{2} + \frac{1}{4} \cdot \frac{x^{-3}}{-3} + c = \boxed{\frac{x^2}{2} - \frac{1}{12x^3} + c}$$

(b) (i) In Δs ABR, CBP

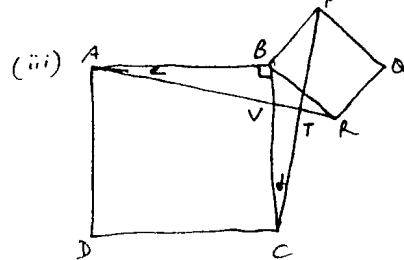
AB = BC, sides of square ABCD

BR = BP, sides of square BPRQ

$\hat{ABR} = \hat{PBC}$, both 90° plus (common) angle ABC
(angle in a square)

$\therefore \Delta ABR \equiv \Delta CBP$, SAS

(ii) both are corresponding angles in congruent Δs in (i)



Let AR meet BC at V & $\angle BAR = \angle$

Then, $\angle BVR = 90^\circ + \angle$, exterior $\angle$ theorem

$\therefore \Delta ABV$
($\angle B = 90^\circ$, angle in a square)

$\therefore \angle VTC = 90^\circ + \angle - \angle VCT$, ext. $\angle$ thm
 $\therefore \Delta VTC$

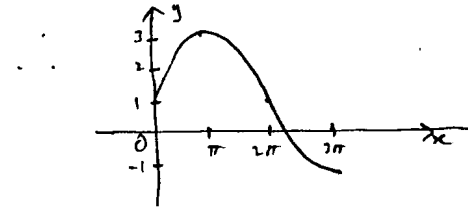
But $\angle VCT = \angle$ (ii)

$\therefore \angle VTC = 90^\circ \Rightarrow AR \perp PC$

Qn 9

(a) (i)

x	0	π	2π	3π
y	1	3	1	-1

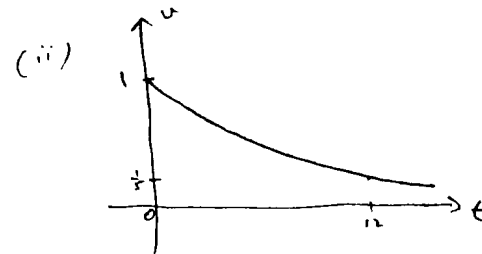


$$(ii) V = \pi \int_0^\pi (2 \sin(\frac{x}{2}) + 1)^2 dx$$

$$(iii) V \approx \pi \cdot \frac{1}{6} \cdot \pi \left[1^2 + 3^2 + 4(2 \sin \frac{\pi}{4} + 1)^2 \right] \text{ m}^3$$

$$= \boxed{54.8 \text{ m}^3}, 1 \text{ d.p.}$$

$$(b) (i) t=0, v = \boxed{1} \text{ m/s}, t=12, v = \frac{1}{\sqrt{25}} \text{ m/s} = \boxed{\frac{1}{5}} \text{ m/s}$$



$$(iii) v = (2t+1)^{-\frac{1}{2}}$$

$$\therefore \ddot{x} = -\frac{1}{2} (2t+1)^{-\frac{3}{2}} \cdot 2 = -\frac{1}{(2t+1)^{3/2}}$$

$$t=12, \ddot{x} = -\frac{1}{25^{3/2}} \text{ m/s}^2 = \boxed{-\frac{1}{125} \text{ m/s}^2}$$

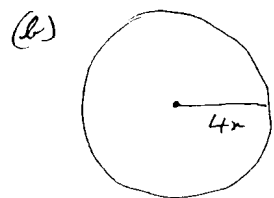
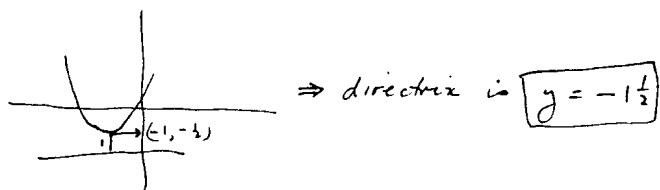
$$(iv) x = \int (2t+1)^{-\frac{1}{2}} dt = \frac{2(2t+1)^{\frac{1}{2}}}{2} + c = \sqrt{2t+1} + c$$

$$t=0, x=0 \Rightarrow 0 = 1+c, c=-1 \quad \therefore x = \boxed{\sqrt{2t+1} - 1}$$

Qn 10

(a) Rewrite as $(x+1)^2 = 4(y+\frac{1}{2})$

$\therefore$ Vertex = $(-1, -\frac{1}{2})$, focal length = 1



(i) Circumference of circle = $2\pi(4x)$
 $= 8\pi x$

$\therefore$ each side of the squares

$$= \frac{200 - 8\pi x}{8} = 25 - \pi x$$

(ii) We must have $4x \geq 0$ and $25 - \pi x \geq 0$

$$\Rightarrow x \geq 0 \quad \text{and} \quad \pi x \leq 25$$

$$\text{ie } x \leq \frac{25}{\pi}$$

$$\therefore 0 \leq x \leq \frac{25}{\pi}$$

$$\begin{aligned} \text{(iii)} \quad A &= \pi(4x)^2 + 2(25 - \pi x)^2 \\ &= 16\pi x^2 + 2(625 - 50\pi x + \pi^2 x^2) \\ &= 16\pi x^2 + 1250 - 100\pi x + 2\pi^2 x^2 \\ &= 2\pi(8 + \pi)x^2 - 100\pi x + 1250 \end{aligned}$$

$$\begin{aligned} \text{(iv)} \quad \frac{dA}{dx} &= 4\pi(8 + \pi)x - 100\pi \\ &= 0 \quad \text{if} \quad x = \frac{100\pi}{4\pi(8 + \pi)} = \frac{25}{8 + \pi} \end{aligned}$$

$$\frac{d^2A}{dx^2} = 4\pi(8 + \pi) > 0 \quad \text{for all } x$$

$\therefore$ curve for A (a parabola) is concave upwards

$\therefore$ least A occurs when $x = \frac{25}{8 + \pi}$

$$\text{(v)} \quad \text{minimum } A = 2\pi(8 + \pi) \frac{625}{(8 + \pi)^2} - 100\pi \cdot \frac{25}{8 + \pi} + 1250$$

$$= \frac{1250\pi}{8 + \pi} - \frac{2500\pi}{8 + \pi} + 1250$$

$$= 1250 - \frac{1250\pi}{8 + \pi}$$

$$= \frac{1250(8 + \pi) - 1250\pi}{8 + \pi} = \frac{10000}{8 + \pi}$$

(vi) From (ii) and (iv), maximum area occurs when $x = 0$ or $x = \frac{25}{\pi}$

The axis of symmetry of the parabola $A = 2\pi(8 + \pi)x^2 - 100\pi x + 1250$

$$\text{is } x = \frac{25}{8 + \pi} \approx 2.24$$

and $\frac{25}{\pi} \approx 7.96$, further from 2.24 than 0

$\Rightarrow$ maximum A occurs when $x = \frac{25}{\pi}$